

Small-Signal Analysis and Optimal Design of External Ramp for Constant On-Time V^2 Control With Multilayer Ceramic Caps

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Abstract—Recently, constant on-time V^2 control, and its variety named constant on-time control, or constant on-time ripple-based control is more and more popular in industry products due to features of high light-load efficiency, simple implementation, and fast transient response. However, subharmonic oscillation occurs when using multilayer ceramic caps due to the lagging phase of the capacitor voltage relative to the inductor current. External ramp compensation is one simple solution to solve the instability issue. However, the characteristics of constant on-time V^2 control with external ramp are not fully understood and no explicit design guidelines for the external ramp are provided. This paper investigates the small-signal characteristics of constant on-time V^2 control with external ramp compensation by providing a factorized, easy-to-use small-signal model. The external ramp is a critical parameter because it directly affects the position and damping of two pairs of double poles. Based on this fact, design guidelines of the external ramp for optimal dynamic performance are provided. Moreover, the effect of duty cycle is investigated. Finally, the small-signal experimental results and load transient performance are presented to verify the small-signal analysis and proposed design guideline.

Index Terms—Constant on-time V^2 , ceramic cap, control-to-output, design guideline, dynamic performance, external ramp, output impedance, ripple-based constant on-time control.

I. INTRODUCTION

THE constant on-time V^2 control architecture [1]–[5], as shown in Fig. 1, has been widely applied in point of load Buck converters mainly due to the following three features: 1) a simple control architecture without a current sensing network and with a simple integrator outer-loop compensator (this

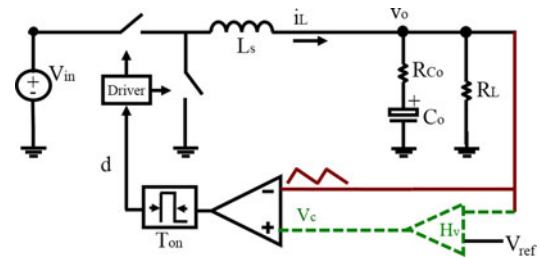


Fig. 1. Structure of constant on-time V^2 control.

outer-loop compensation is shown as a dashed line in Fig. 1 since in many applications, the integrator can be further eliminated, which is called constant on-time control [3] or ripple-based constant on-time control [10]), 2) fast-load transient characteristics with direct output voltage feedback, and 3) good light-load efficiency with a constant on-time structure.

This structure is simple and elegant only when the ESR ripple is sufficient compared with the capacitor voltage ripple, e.g., when an OSCON cap is used. However, in many portable electronics applications such as cell phones and cameras, low-ESR capacitors such as multilayer ceramic caps are preferred due to small size, low cost, and small output voltage ripple. In the case where ceramic caps are used, V^2 control or ripple-based constant on-time control suffers from the subharmonic oscillation due to the lagging phase of the capacitor voltage relative to the inductor current [10], [11]. External ramp compensation is a well-known solution to eliminate subharmonic oscillation in peak current-mode control, and this strategy can be employed to eliminate subharmonic oscillation in constant on-time V^2 control as well [11], [12]. For constant on-time control, ramp compensation can be fulfilled by adding a ramp with a fixed level (zero for simplicity) during on-time interval and a fixed slope during off-time interval. Compared with other methods [13]–[20], external ramp compensation is very simple to be implemented inside the control chip and no additional current sensing circuit is required.

To predict the instability problem for constant on-time V^2 control and provide design guidelines for ramp compensation, an accurate small-signal model is necessary. In V^2 or ripple-based control implementation, the nonlinear pulse width modulation (PWM) modulator becomes much more complicated, because not only the inductor current information is fed back to the modulator, but also the capacitor voltage ripple information. As a result, the modeling of constant on-time V^2 control

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becomes a challenge due to the complexity of the PWM modulator. It has been proved that subharmonic oscillation is not caused by the inductor current feedback [21]–[23] but by the phase delay due to large capacitor voltage ripple [10], [11]. The models used in [6]–[9] fail to predict subharmonic oscillation since these models are extensions of Ridley's model [22], which is based on constant-frequency discrete-time analysis and does not consider the influence of the capacitor voltage ripple. In [10], the Krylov–Bogoliubov–Mitropolsky ripple estimation technique is used to predict the influence of the capacitor voltage ripple and the effect of the external ramp; however, this technique is too complex for practical use. Besides, the conclusion shown in [10] is only applicable for constant frequency V^2 control, and no small-signal model and design analysis is presented for constant on-time V^2 control. In [24] and [25], a new subharmonic analysis based on time-domain waveforms is presented. The stability point can be predicted and the minimum external ramp required to stabilize the circuit can be calculated. However, only the absolute stability point can be predicted, and no relative margin can be controlled for design purposes. Besides, design tradeoffs between the external ramp and the load transient performance is still unclear. In [11], the describing function (DF) method is used to model the nonlinear PWM modulator for V^2 control to obtain transfer functions of control-to-output and output impedance. This method considers all the nonlinearities due to inductor current feedback and capacitor voltage feedback and is accurate even beyond half of switching frequency. The small-signal model of constant on-time V^2 control with external ramp compensation is also derived in [11]. However, the transfer function is in a nonfactorized complicated form and the small-signal characteristic has not been explored in detail. Besides, no explicit design guideline for the external ramp is provided.

The research goal of this paper is to investigate the small-signal characteristic for constant on-time V^2 control with external ramp compensation based on the presented factorized small-signal model, and provide a design guideline of the external ramp for optimal transient performance. The remainder of this paper is organized as follows. Section II reviews the previous small-signal model with external ramp compensation. In Section III, the factorized small-signal model is presented and pole-zero movements are identified for both control-to-output transfer function and output impedance. Based on that, the optimal design guideline for external ramp is provided from the stability margin and transient performance point of view. In Section IV, the effect of the duty cycle is investigated. In Section V, simulation and experimental verifications are presented to verify the small-signal characteristic and proposed external ramp design guideline. Finally, a summary is given in Section VI.

II. PREVIOUS SMALL-SIGNAL MODEL WITH EXTERNAL RAMP COMPENSATION

Fig. 2(a) shows the circuit diagram of constant on-time V^2 control with external ramp compensation. As shown in Fig. 2(a), the external ramp starts to build up at the end of the on-time period, and resets at the beginning of the on-time period in every

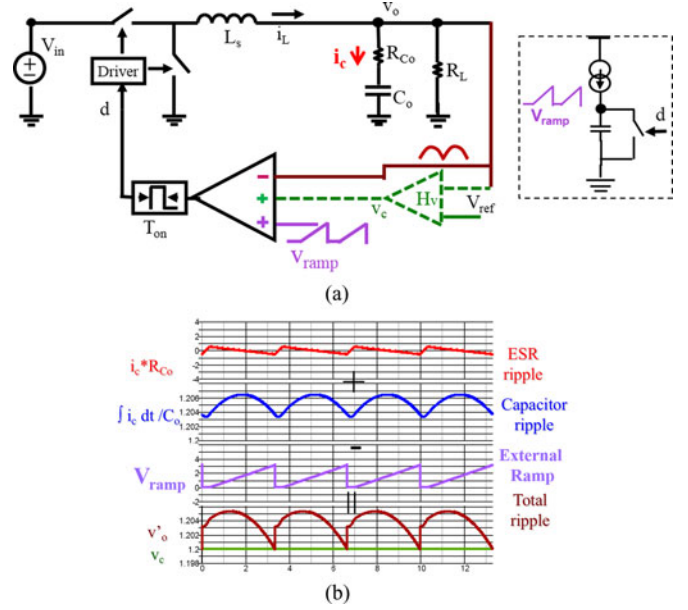


Fig. 2. Constant on-time V^2 control with external ramp compensation. (a) Circuit diagram. (b) Steady-state waveforms.

switching cycle. The external ramp can be easily implemented inside controller IC with a current source charging a capacitor and a PWM signal controlled switch. Fig. 2(b) illustrates the steady-state waveforms. Although the output voltage ripple is dominated by the capacitor voltage ripple, with the help of the external ramp which is in the same phase as the ESR ripple, the delay effect caused by the capacitor voltage ripple is reduced and the circuit becomes stable.

The small-signal control-to-output transfer function is provided in [11] based on the timing-domain DF method, and it is accurate even beyond switching frequency. Based on Pade approximation, a simplified polynomial model accurate up to switching frequency is shown as follows:

$$\frac{v_o(s)}{v_c(s)} \approx \frac{1}{(1 + \frac{s}{Q_1\omega_1} + \frac{s^2}{\omega_1^2})} \cdot \frac{(1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2})(R_{Co}C_o s + 1)}{(1 + \frac{s}{Q_3\omega_3} + \frac{s^2}{\omega_3^2})(1 + \frac{s}{Q_1\omega_1} + \frac{s^2}{\omega_1^2}) + \frac{s_e}{s_f} R_{Co}C_o T_{sw} s^2} \quad (1)$$

where T_{sw} is the switching period, R_{Co} is the ESR of the output capacitors, C_o is the capacitance of the output capacitors, $\omega_1 = \pi/T_{on}$, $Q_1 = 2/\pi$, $\omega_2 = \pi/T_{sw}$, $Q_2 = 2/\pi$, $Q_3 = T_{sw}/[R_{Co}C_o - T_{on}/2]\pi$, T_{on} is the on-time, s_e is the slope of the external ramp during off-time interval, $s_f = R_{Co}V_o/L_s$, and L_s is the inductance of the inductor.

For most applications, the switching frequency is around several hundreds of kilohertz and the zero caused by the ESR of the ceramic cap is usually higher than 1 MHz and can, therefore, be neglected. If the duty ratio is small, the double pole caused by on-time is far away from switching frequency and can also be neglected. Therefore, the transfer function can be

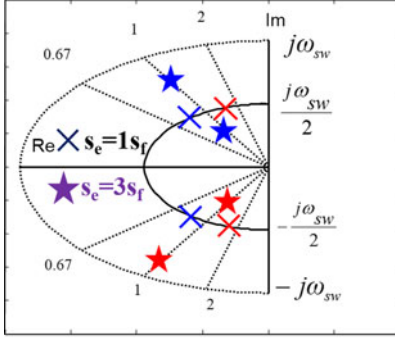


Fig. 3. Locations of two pairs of double poles with two different external ramps.

further simplified as follows:

$$\frac{v_o(s)}{v_c(s)} \approx \frac{(1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2})}{(1 + \frac{s}{Q_3\omega_2} + \frac{s^2}{\omega_2^2})(1 + \frac{s}{Q_1\omega_2} + \frac{s^2}{\omega_2^2}) + \frac{s_e}{s_f} R_{Co} C_o T_{sw} \cdot s^2} \quad (2)$$

From (2), the denominator is fourth order and the poles are affected by the external ramp using the numerical solution. The characteristics of the poles are very strange and not fully understood. Fig. 3 shows one example with the following circuit parameters: $V_{in} = 12$ V, $V_o = 1.2$ V, $D = 0.1$, $F_{sw} = 300$ kHz, $C_o = 100$ μ F * 8, $R_{Co} = 1.4$ m Ω /8, $L_s = 600$ nH, and angular frequency $\omega_{sw} = 2\pi f_{sw}$. With two different external ramp designs, the positions and the damping of the poles are different. However, using the numerical solution, the poles become very difficult to identify and the relations between the locations of the poles, the quality factors, and the circuit parameters are not clear. In addition, the numerical solution is very time consuming and no design guideline is provided.

One of the benefits of V^2 control is fast transient response. Therefore, it is meaningful to investigate the output impedance. The transfer function of the output impedance is also derived in [11] and shown as follows:

$$Z_o(s) \approx \left(\frac{1}{1 + \frac{s}{Q_1\omega_1} + \frac{s^2}{\omega_1^2}} \times \frac{(1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2})}{(1 + \frac{s}{Q_3\omega_2} + \frac{s^2}{\omega_2^2})(1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2}) + \frac{s_e}{s_f} R_{Co} C_o T_{sw} \cdot s^2} - 1 \right) \times \left(R_{Co} + \frac{1}{C_o s} \right) \quad (3)$$

For the same reason, the characteristics of the output impedance are not very clear since the denominator is not factorized. Factorized model and pole-zero identification are necessary for analysis and design purposes.

III. FACTORIZED SMALL-SIGNAL MODEL AND EXTERNAL RAMP DESIGN GUIDELINE

Based on the analysis in Section II, it is meaningful and worthwhile to identify the pole-zero movements by factorizing the fourth-order denominator. The transfer function shown in (2) can be factorized in the following form:

$$\frac{v_o(s)}{v_c(s)} \approx \frac{(1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2})}{(1 + \frac{s}{Q_{e1}(\omega_2/a)} + \frac{s^2}{(a\omega_2)^2})} \cdot \frac{1}{(1 + \frac{s}{Q_{e2}(\omega_2/a)} + \frac{s^2}{(\omega_2/a)^2})} \quad (4)$$

where a , Q_{e1} , and Q_{e2} are all real numbers. The factorized results are divided into two regions based on different locations of the double poles due to different amplitudes of the external ramp, as explained in following.

Region I: for a relatively small external ramp which met the following equation:

$$s_e \leq \frac{(1 - 2\alpha + D)^2}{16\alpha} s_f \quad (5)$$

where α is defined as follows:

$$\alpha = \frac{R_{Co} C_o}{T_{sw}} \quad (6)$$

The physical meaning of α is that α represents the relative strength of the current feedback: the larger the α value, the larger the ESR ripple when compared with the capacitor voltage ripple and therefore the stronger the current information feedback. For the example using 1.4 m Ω /100 μ F ceramic caps and with 300-kHz switching frequency, α is around 0.04. For a detailed analysis and discussion about α , please refer to [26].

The expressions of a , Q_{e1} , and Q_{e2} in (4) are shown as follows:

$$a = 1$$

$$Q_{e1} = \frac{4}{\pi} \frac{1}{1 + 2\alpha - D + \sqrt{(1 - 2\alpha + D)^2 - 16\alpha s_e/s_f}}$$

$$Q_{e2} = \frac{4}{\pi} \frac{1}{1 + 2\alpha - D - \sqrt{(1 - 2\alpha + D)^2 - 16\alpha s_e/s_f}} \quad (7)$$

In this region, $a = 1$ means that both pairs of double poles are located at half the switching frequency. The external ramp changes Q_{e1} and Q_{e2} , as shown in (7). For the purpose of explanation, the same circuit parameters as Fig. 3 are used as an example: $V_{in} = 12$ V; $V_o = 1.2$ V; $D = 0.1$; $F_{sw} = 300$ kHz; Ceramic Caps: $R_{Co} = 1.4$ m Ω /8, $C_o = 100$ μ F * 8, $L_s = 600$ nH. The pole-zero movement with increasing external ramp is shown in Fig. 4.

When $S_e = 0$, which is indicated by point A in Fig. 4, $Q_{e1} = 2/\pi$, $Q_{e2} = T_{sw}/([R_{Co} C_o - T_{on}/2]\pi) = -39.6$, the blue pair of double poles is located on the left half-plane and cancels the double zero, which is located on the same position. The red pair of double poles is located on the right half-plane since the quality factor is negative, which means that the system is unstable without external ramp compensation. The stability criterion

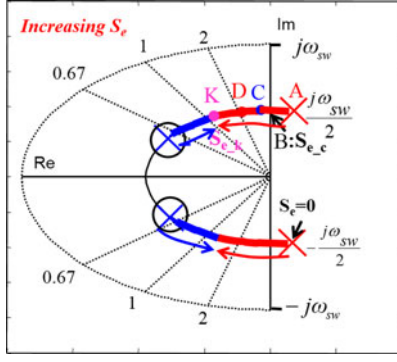


Fig. 4. Pole-zero map of control-to-output transfer function with increasing s_e in region I.

$R_{Co} C_o > T_{on}/2$ is validated with simulation and experimental results in [11].

When S_e increases, the double zero does not move while the quality factors of the two pairs of double poles are affected. As (7) indicates, Q_{e1} remains positive and increases the value, while Q_{e2} changes from negative value to positive. In Fig. 4, as S_e increases, the left half-plane pair of double poles moves to the right and the right half-plane pair moves to the left. At point B, the second pair of double poles enters into the left half-plane, which means that the system becomes stable. The stability criterion for the external ramp to stabilize the system can be solved as follows:

$$\left(2\frac{s_e}{s_f} + 1\right)\alpha - \frac{D}{2} > 0 \text{ or}$$

$$s_e > s_{e-c} = \left(\frac{D}{4\alpha} - \frac{1}{2}\right)s_f = \frac{V_o T_{sw}}{4L_s C_o}(D - 2\alpha). \quad (8)$$

For the specific case of study, $D = 0.1$, $\alpha = 0.04$, the system will be stable when $S_e > S_{e-c} = 0.1S_f$. The stability criterion is the same as in [25], where it is calculated based on discrete-time analysis and is verified with simulation and experiment. After point B, this pair of double poles moves further into the left-half-plane, through points C and D, until at point K, these two pairs of double poles are located at the same position. Point K is called the key point as this point is very important for ramp design. This point is actually the border of regions I and II. The magnitude of the external ramp to reach point K is shown in (9) and the quality factor at point K is shown in (10) as follows:

$$\text{Key point: } s_{e-K} = \frac{(1 - 2\alpha + D)^2}{16\alpha} s_f \quad (9)$$

$$\text{Key point: } Q_{e1} = Q_{e2} = Q_{e-K} = \frac{4}{\pi} \frac{1}{1 + 2\alpha - D}. \quad (10)$$

For the case of study with $D = 0.1$ and $\alpha = 0.04$, $S_{e-K} = 1.5S_f$ and $Q_{e-K} = 1.3$. Fig. 5 compares Bode plots of control-to-output transfer function with different external ramps in this region. It is clearly shown that with a larger external ramp, the peaking of the gain curve at half the switching frequency is smaller since Q_{e2} is smaller. From a design point of view, key point K is the best point in this region, since the system has the highest damping.

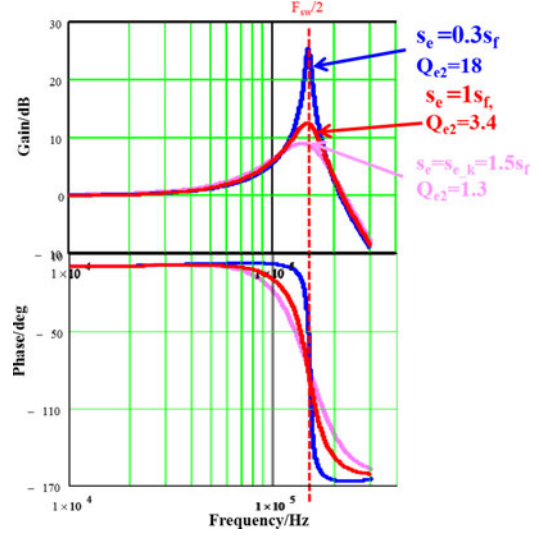


Fig. 5. Bode plots comparison of control-to-output transfer function in region I.

From a physical point of view, since the external ramp is in phase with the ESR ripple, by adding an external ramp, the weight of the capacitor voltage ripple in PWM modulation is reduced. Therefore, the delay effect is reduced and dynamic performance improves.

Region II: for relatively large external ramp which met the following equation:

$$s_e \geq \frac{(1 - 2\alpha + D)^2}{16\alpha} s_f. \quad (11)$$

The expression of a in (4) is shown as follows:

$$a = \frac{\sqrt{4 + Y} + \sqrt{Y}}{2} > 1 \quad (12)$$

$$Y = \frac{\pi^2}{4} \left[\left(2\frac{s_e}{s_f} + 1\right)\alpha - \frac{D}{2} \right] - 2 + \frac{1}{2}$$

$$\cdot \sqrt{\left(\frac{\pi^2}{2} \left[\left(2\frac{s_e}{s_f} + 1\right)\alpha - \frac{D}{2} \right] + 4 \right)^2 - \pi^2 (1 + 2\alpha - D)^2}. \quad (13)$$

The expressions of Q_{e1} and Q_{e2} in (4) are shown as follows:

$$Q_{e1} = Q_{e2} = \frac{2}{\pi} \frac{1}{1 + 2\alpha - D} \left(a + \frac{1}{a} \right) \quad (14)$$

As external ramp increases, $a > 1$ in (12) means that the two pairs of double poles are separated and are not located at half the switching frequency any more. One pair of double poles moves to a higher frequency, while the other pair of double poles moves to a lower frequency. The interesting phenomenon is that these two pairs of double poles have the same quality factor in this region which is shown in (14). Comparing (7), (10), and (14), it is found that at the key point Q_{e2} reaches its minimum value. On further increase in the external ramp after the key point K, the two pairs of double poles are separated, while the quality factor increases a little bit. The pole-zero maps with increasing external ramp in this region are shown in Fig. 6.

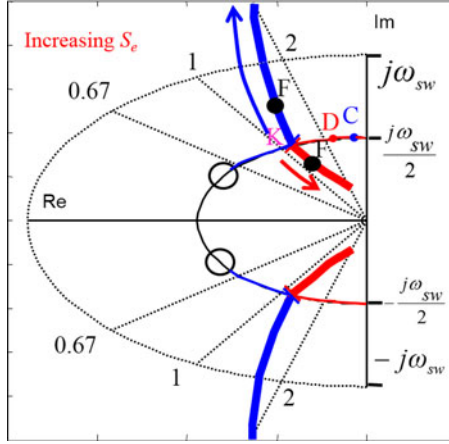


Fig. 6. Pole-zero map of control-to-output transfer function with increasing s_e in region II.

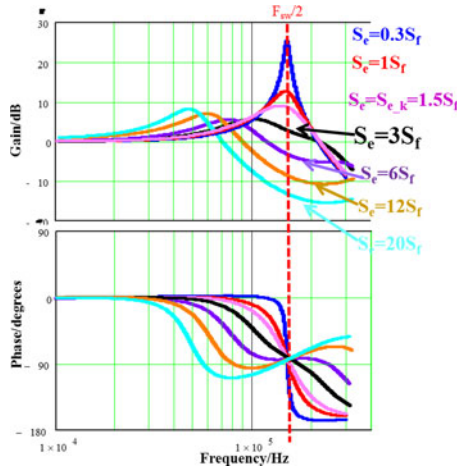


Fig. 7. Bode plots comparison of control-to-output transfer function in regions I and II.

Fig. 7 shows a comparison of Bode plots of the control-to-output transfer function with different external ramps, including both regions I and II. On one hand, separating the two pairs of poles is beneficial for the stability margin since it pushes one pair of double poles to a frequency higher than half the switching frequency and, therefore, reduces the peaking of the gain curve. This point is verified in Fig. 7, which shows that the peaking of gain curve with $s_e = 3S_f$ and $s_e = 6S_f$ is less than the peaking of gain curve at the key point with $s_e = 1.5S_f$. On the other hand, separating the two pairs of double poles too far apart is not preferred due to the following two reasons: first, the lower frequency double pole will cause a phase drop in the lower frequency range and this low-frequency double pole slows the transient performance. Second, the peaking of the double poles will be larger, which can be seen in (14): as variable a increases, Q_{e1} and Q_{e2} both increase. This point is also verified in Fig. 7, which shows that the peaking of the gain curve with $s_e = 12S_f$ and $s_e = 20S_f$ is larger than the peaking of the gain curve with $s_e = 3S_f$ and $s_e = 6S_f$.

From a physical point of view, in this region, too much external ramp will gradually change constant on-time V^2 control

into constant on-time voltage mode control, as the large external ramp also reduces the effect of the feedback current information. Note that the double zero which is located at half of the switching frequency and with constant Q equals to $2/\pi$ brings phase boost at high frequency. This agrees with the constant on-time modulation model presented in [27], where a phase leading property for constant on-time modulator is observed. With an increasing external ramp, the modulator gain reduces, and therefore, the bandwidth reduces gradually, resulting in a reduction of the phase leading at crossover frequency. Therefore, the phase margin decreases and the quality factor increases, which explains the low-frequency moving pole and increasing peaking on the gain curve.

For the output impedance, with the denominator factorized, (3) can be simplified and rewritten as follows:

$$Z_o(s) \approx \frac{-b \cdot s}{\left(1 + \frac{s}{Q_{e1} \cdot (a\omega_2)} + \frac{s^2}{(a\omega_2)^2}\right)} \cdot \frac{1}{\left(1 + \frac{s}{Q_{e2} \cdot (\omega_2/a)} + \frac{s^2}{(\omega_2/a)^2}\right)} \cdot \frac{R_{Co}C_o s + 1}{1 + \frac{s}{Q_1\omega_1} + \frac{s^2}{\omega_1^2}},$$

where $b = \frac{s_e}{s_f} R_{Co} T_{sw}$

$$+ \frac{T_{sw}^2}{C_o} \left[\frac{1 + D^2}{\pi^2} + \frac{D}{2} \left(\frac{R_{Co}C_o}{T_{sw}} - \frac{D}{2} \right) \right]. \quad (15)$$

For a small duty cycle application, with the ESR zero of ceramic caps much higher than switching frequency, (15) can be simplified as follows:

$$Z_o(s) \approx \frac{-b \cdot s}{\left(1 + \frac{s}{Q_{e1} \cdot (a\omega_2)} + \frac{s^2}{(a\omega_2)^2}\right)} \cdot \frac{1}{\left(1 + \frac{s}{Q_{e2} \cdot (\omega_2/a)} + \frac{s^2}{(\omega_2/a)^2}\right)}$$

where $b = \frac{s_e}{s_f} R_{Co} T_{sw} + \frac{T_{sw}^2}{C_o} \frac{1}{\pi^2}$. (16)

From (16), the output impedance has a zero at origin, and its coefficient is related to the external ramp. The poles of the output impedance are the same as the control-to-output transfer function. The comparisons of output impedance with different external ramp compensations are shown in Fig. 8.

As can be seen, the output impedance is very small in the whole frequency region and this explains why the transient response is very fast. However, the magnitude and shape of the output impedance are affected by the external ramp. When the external ramp is too small, the large peaking occurs in the gain plot. When the external ramp is too large, large peaking also occurs since the quality factor of the low-frequency double pole is increasing. Besides, the output impedance in the low frequency increases since the coefficient b in (16) is proportional to the external ramp. Therefore, either too small or too large external ramp is detrimental to the load transient performance.

From the above analysis, the choice of external ramp is critical for the dynamic performance. On one hand, the external ramp

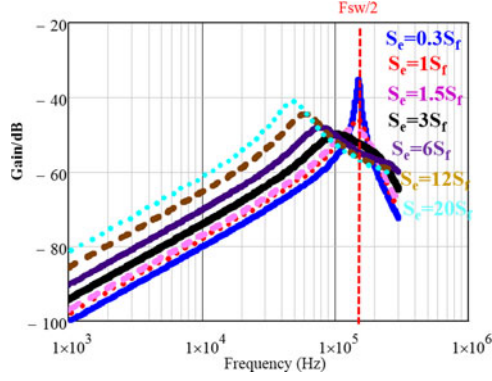


Fig. 8. Bode plots comparison of output impedance in regions I and II.

should be designed larger than S_{e_k} which is the key point K. On the other hand, the external ramp should not be too large to avoid the lower dominant double pole with large quality factor. The recommended external ramp is around $S_e = 2S_{e_k}$. Based on the factorized small-signal model, the proposed external ramp can be calculated as follows:

$$\begin{aligned} \text{Preferred } s_e: s_e &\approx 2s_{e_k} = \frac{(1 - 2\alpha + D)^2}{8\alpha} s_f \\ &= \frac{V_o T_{sw}}{8L_s C_o} (1 - 2\alpha + D)^2. \end{aligned} \quad (17)$$

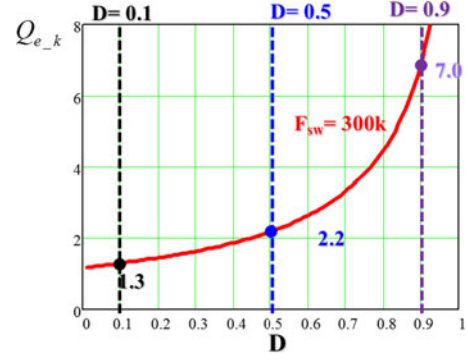
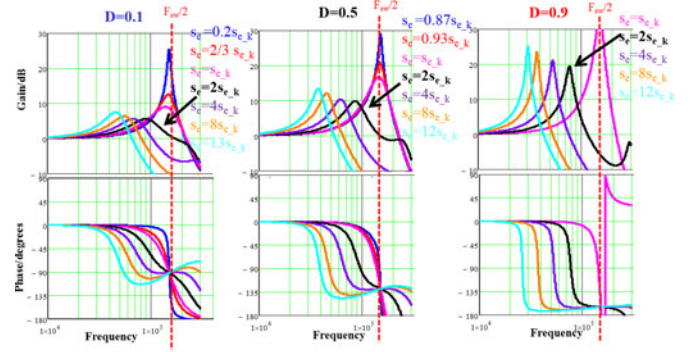
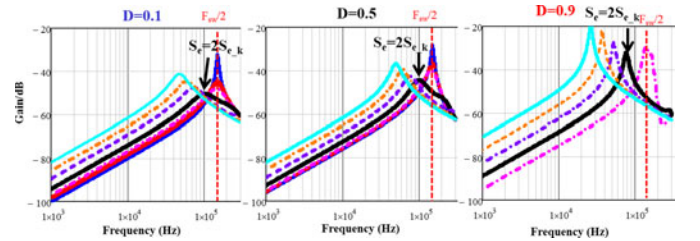
In this example, $S_e = 3S_f$ is preferred, which is the black curve shown in Figs. 7 and 8. Small-signal verification and load transient performance verification are provided in Section V.

IV. EFFECT OF DUTY CYCLE

For small duty cycle, the double pole located at $\omega_1 = \pi/T_{on}$ is far away from f_{sw} ; therefore, the effect is very small and can be neglected. When duty cycle becomes larger, this pair of double pole is closer to switching frequency and cannot be neglected any more. For example, when $D = 0.1$, $\omega_1 = 5\omega_{sw} \gg \omega_{sw}$. When $D = 0.75$, $\omega_1 = 2/3\omega_{sw}$, the double pole is very close to $\frac{1}{2}f_{sw}$ and will cause additional phase delay at high frequency. The factorized control-to-output transfer function for general case is shown as follows:

$$\begin{aligned} \frac{v_o(s)}{v_c(s)} &\approx \frac{1}{1 + \frac{s}{Q_1\omega_1} + \frac{s^2}{\omega_1^2}} \cdot \frac{(1 + \frac{s}{Q_2\omega_2} + \frac{s^2}{\omega_2^2})}{(1 + \frac{s}{Q_{e1}(a\omega_2)} + \frac{s^2}{(a\omega_2)^2})} \\ &\cdot \frac{1}{(1 + \frac{s}{Q_{e2}(\omega_2/a)} + \frac{s^2}{(\omega_2/a)^2})}. \end{aligned} \quad (18)$$

The first pair of double pole is well damped with $Q_1 = 2/\pi$, which means that it does not affect the peaking and will not cause stability problem. However, duty cycle affects the quality factor of the other two pairs of double poles. Referring to (10), the quality factor Q_{e2} at key point is related to duty cycle and α . For 1.4-mΩ/100-μF ceramic caps, with 300-kHz switching frequency, α is only 0.04. The relation between Q_{e_k} and D is plotted as Fig. 9. It is obvious that with an increasing duty cycle, the minimum quality factor Q_{e_k} is

Fig. 9. Relations between Q_{e_k} and D with $F_{sw} = 300$ kHz.Fig. 10. Bode plots comparisons of control-to-output transfer function with different S_e for $D = 0.1, 0.5$, and 0.9 .Fig. 11. Bode plots comparisons of output impedance with different S_e between $D = 0.1, 0.5$, and 0.9 .

nearly 7 when D equals 0.9, which indicates that the dynamic performance with large duty cycle is very bad.

The Bode plots of the control-to-output transfer function [see (18)] and output impedance [see (15)] when duty cycle is 0.1, 0.5, and 0.9 are shown in Figs. 10 and 11, respectively. Several points can be concluded from Fig. 10: first, for $D = 0.5$ and $D = 0.9$ cases, the effect of the external ramp is very similar with the $D = 0.1$ case. In region I, increasing the external ramp reduces the peaking at half the switching frequency; in region II, increasing the external ramp separates the two pairs of double poles and also increases the quality factor. It is also obvious that the preferred external ramps in these cases are also around $2S_{e_k}$, which is the same as $D = 0.1$ case. Second, comparing the gain plots of $D = 0.1$, $D = 0.5$, and $D = 0.9$ cases, the peaking increases with increasing duty cycle. The smallest peaking is around 10 dB with $D = 0.5$ and around 20 dB with $D = 0.9$. From the output impedance shown in Fig. 11, it is also

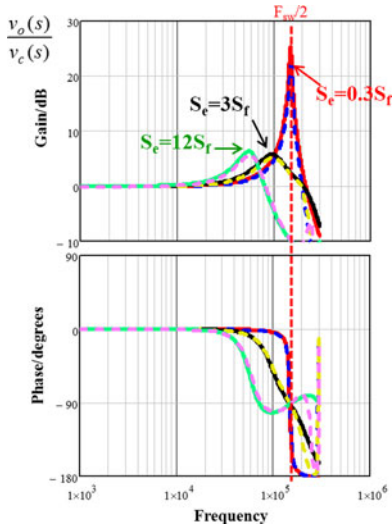


Fig. 12. Bode plots of control-to-output transfer function between model and simulation for different external ramp: Solid line—Model; Dashed line—SIMPLIS simulation.

shown that the preferred external ramp in each case is around $2S_{e,k}$. When the duty cycle is larger, the peaking at the double poles is larger and the low frequency magnitude also increases.

Therefore, designing the external ramp around $2S_{e,k}$ is recommended from stability and dynamic performance point of view. The magnitude of the ramp is shown in (17). Since α is very small for ceramic caps and can be neglected, the preferred external ramp is only related with the power stage parameter V_o , L_s , C_o , T_{sw} , and D . When duty cycle is larger, the dynamic performance is worse. The conclusion is consistent with the results in [27] where it is found that the phase leading property of the constant on-time modulator is related to off-time: larger D means smaller off-time, which leads to smaller phase leading. Therefore, the phase margin inside the loop is smaller and the dynamic performance is worse. Generally speaking, with a large duty cycle (above 50%) application, an external ramp solution cannot obtain a well-damped performance for ceramic caps.

V. SIMULATION AND EXPERIMENTAL VERIFICATIONS

In [11] and [12], the SIMPLIS simulation tool is used to verify the small-signal model for constant on-time V^2 control with external ramp compensation. To verify the analysis and proposed design guideline in this paper, Fig. 12 shows the comparison between the model and SIMPLIS simulation for the control-to-output transfer function with different external ramp designs, while Fig. 13 shows the comparison for output impedance. The circuit parameters are as follows: $D = 0.1$, $F_{sw} = 300$ kHz, $C_o = 100 \mu\text{F} \cdot 8$, $R_{Co} = 1.4 \text{ m}\Omega$, and $L_s = 600 \text{ nH}$. The figures clearly show that the factorized small-signal model is accurate at half of switching frequency and is useful up to switching frequency. Moreover, the results provided in [11] and [12] are for the small duty cycle case only. To verify the small-signal analysis with different duty cycles, the small-signal model and SIMPLIS simulation for different duty cycles are compared in Fig. 14. The switching frequency in this case is 300 kHz, and the

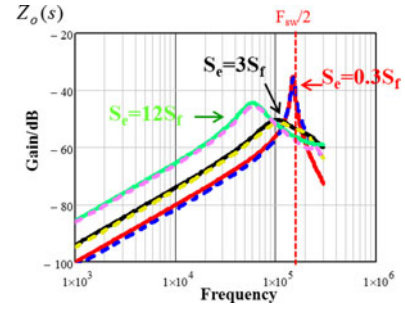


Fig. 13. Bode plots of output impedance between model and simulation for different external ramp: Solid line—Model; Dashed line—SIMPLIS simulation.

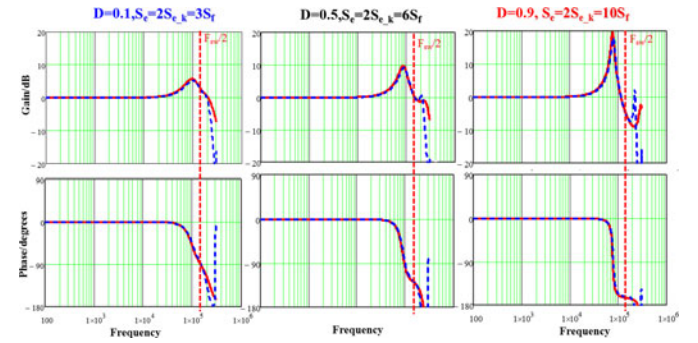


Fig. 14. Bode plots of control-to-output transfer function between model and simulation for different duty cycles: Solid line—Model; Dashed line—SIMPLIS simulation.

external ramp in each case is designed at around $2S_{e,k}$, which is the best choice from previous analysis. It is clear that the small-signal models are accurate at half of switching frequency and up to switching frequency. With increasing duty cycle, the peaking is also increasing. For $D = 0.9$ case, the peaking is around 20 dB, which indicates very small stability margin and bad dynamic performance.

To verify the small-signal characteristic and proposed external ramp design guideline, a prototype of a constant on-time buck converter is built by using discrete components based on controller LM25011A from Texas Instruments. Fig. 15 shows the circuit diagram, and the detailed specification is shown as follows: $V_o = 3.3$ V; F_{sw} is around 300 kHz; $L_s = 8 \mu\text{H}$. The ceramic cap GRM31CR60J107ME39L (100 $\mu\text{F}/6.3$ V) manufactured by Murata is used as output cap. From the datasheet, the actual capacitance is 40 μF considering the dc bias influence and the ESR at 300 kHz is 2 m Ω . Input voltage can be varied from 6 to 35 V, which corresponds to a duty cycle range from 0.1 to 0.5. Feedback resistor R_{FB1} is 4 k Ω and R_{FB2} is 1.1 k Ω . The feedback gain needs to be taken into account in the calculation of S_f and S_e . In the experiments, definition of S_f is $R_{FB1}/(R_{FB1} + R_{FB2}) \cdot R_{Co} \cdot V_o/L_s$. An on-board load transient circuit is built and the small-signal control-to-output transfer function is measured based on the network analyzer Agilent 4395A.

According to (8), the value of external ramp to stabilize the circuit is $2.8S_f$ or 1.75 mV/ μs . To verify the stability criterion, Fig. 16 shows a steady-state waveform with an external ramp $S_e = 1.3$ mV/ μs at 12-V input voltage. As expected from (8),

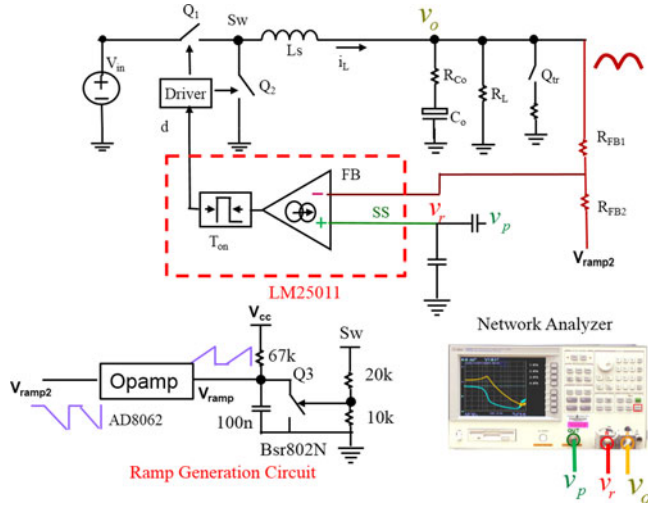
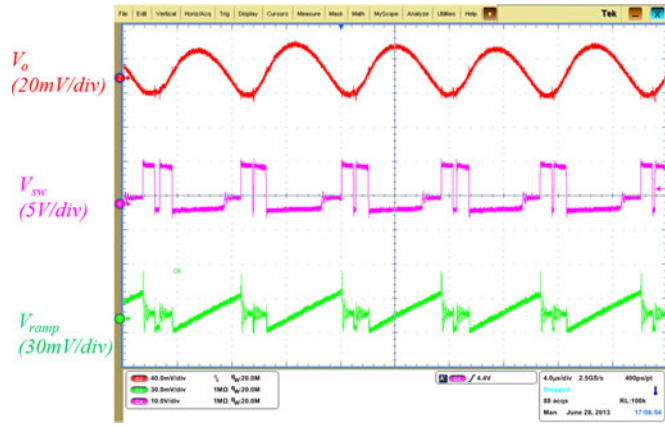
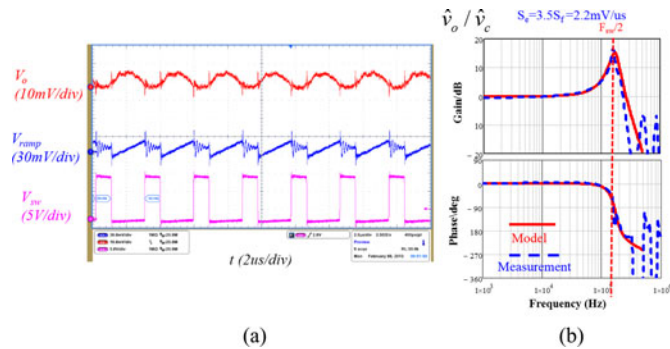
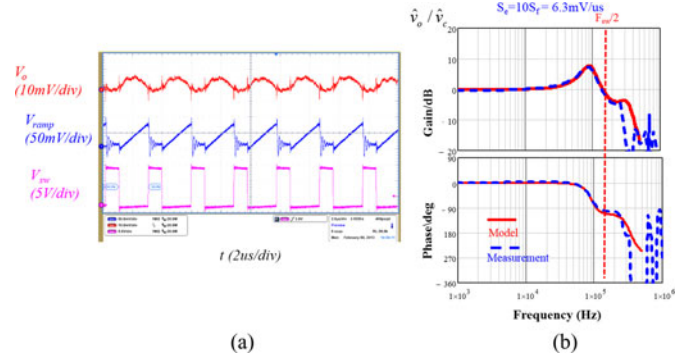
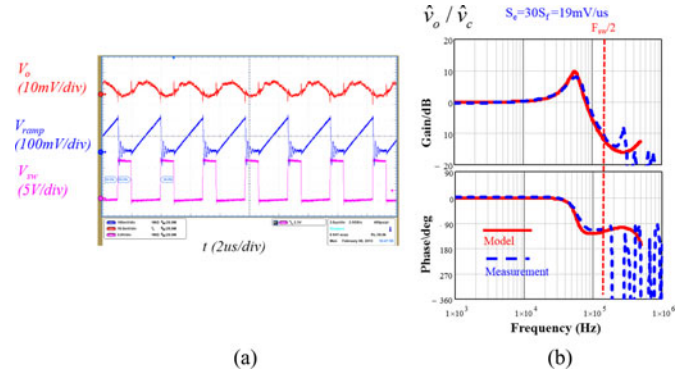


Fig. 15. Circuit diagram of the prototype for experimental verification.

Fig. 16. Subharmonic oscillation for ceramic cap with insufficient external ramp compensation $S_e = 1.3 \text{ mV}/\mu\text{s}$.Fig. 17. (a) Steady-state waveform. (b) Small-signal control-to-output transfer function with $S_e = 3.5S_f = 2.2 \text{ mV}/\mu\text{s}$.

subharmonic oscillation still exists, which is undesired since it increases output voltage ripple and inductor current ripple and reduces the efficiency of the circuit.

Fig. 17(a) shows a steady-state waveform with $S_e = 3.5S_f = 2.2 \text{ mV}/\mu\text{s}$. As expected from (8), the circuit is stabilized. V_{ramp2} signal represents the amplitude of external ramp. Note that V_{ramp2} after the voltage divider gain $R_{fb2}/(R_{fb1} + R_{fb2})$

Fig. 18. (a) Steady-state waveform. (b) Small-signal control-to-output transfer function with $S_e = 10S_f = 6.3 \text{ mV}/\mu\text{s}$.Fig. 19. (a) Steady-state waveform. (b) Small-signal control-to-output transfer function with $S_e = 30S_f = 19 \text{ mV}/\mu\text{s}$.

is the external ramp seen at the input of the modulator. In the experiment, the gain is around 1/5. Therefore, the ramp amplitude should be $2.2 \text{ mV}/\mu\text{s} \times 2 \mu\text{s} \times 5 = 22 \text{ mV}$, which agrees well with the result shown in Fig. 17(a). However, the external ramp in this case is located in region I and is too small from a damping point of view. According to (9), the external ramp at key point in this case is $4S_f$ or $2.5 \text{ mV}/\mu\text{s}$. Fig. 17(b) shows the small-signal control-to-output transfer function. The measurement result shows that the two pairs of double poles in this case are both at $\frac{1}{2} f_{sw}$ with a large quality factor. The measurement result agrees very well with the small-signal model up to switching frequency.

Fig. 18(a) shows a steady-state waveform with $S_e = 10S_f = 6.3 \text{ mV}/\mu\text{s}$. This is the preferred external ramp according to (17). Note that the scale of V_{ramp} is $50 \text{ mV}/\mu\text{s}$, compared with $30 \text{ mV}/\mu\text{s}$ in the previous case. Fig. 18(b) shows a comparison of the small-signal model with measurement result. The results match very well. From the result, we can see that the two pairs of double poles separate, one to lower frequency and the other to higher frequency, which verify the existence of region II. Compared with Fig. 17(b), the peaking reduces which indicates a better dynamic performance.

Fig. 19(a) shows a steady-state waveform with $S_e = 30S_f = 19 \text{ mV}/\mu\text{s}$. The external ramp is too large in this case. Note that the scale of V_{ramp} is $100 \text{ mV}/\mu\text{s}$, compared with $50 \text{ mV}/\mu\text{s}$ in the previous case. Fig. 19(b) shows a comparison of the small-signal model with measurement result. The results

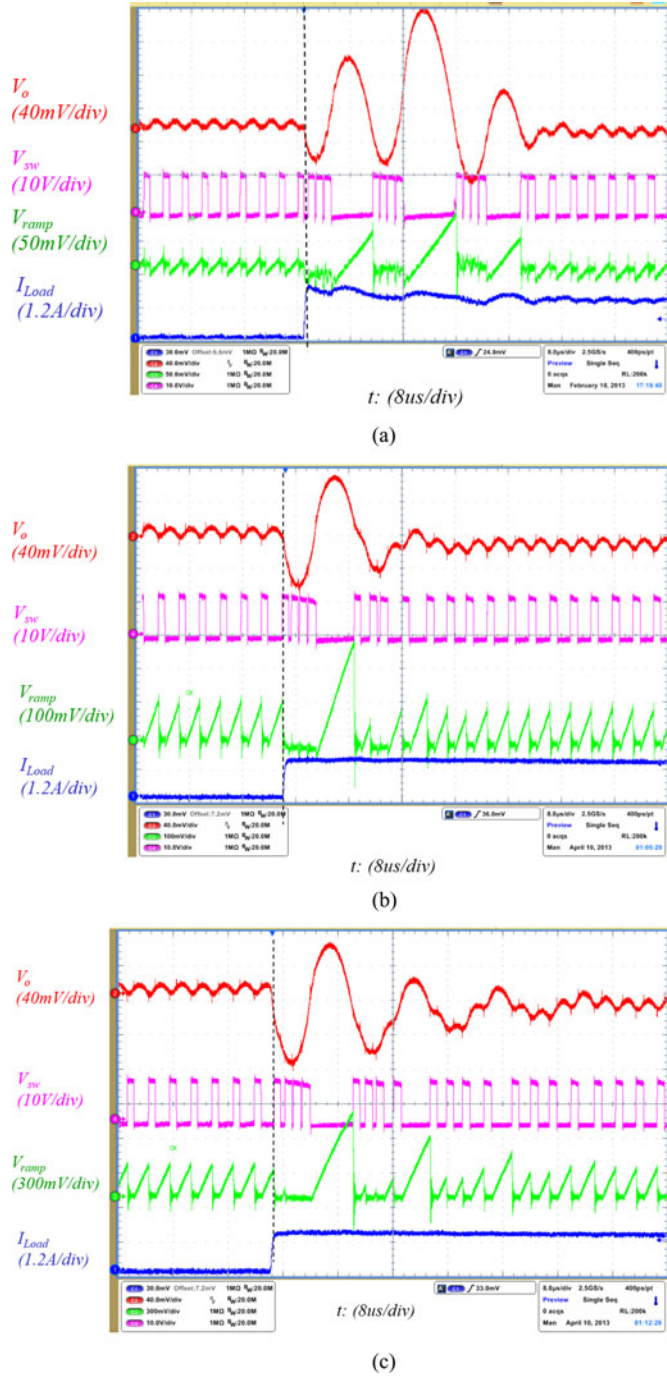


Fig. 20. Load step-up transient performance comparison for 0.5–2 A load step. (a) $S_e = 3.5S_f = 2.2 \text{ mV}/\mu\text{s}$. (b) Worst case for $S_e = 10S_f = 6.3 \text{ mV}/\mu\text{s}$. (c) Worst case for $S_e = 30S_f = 19 \text{ mV}/\mu\text{s}$.

match well enough. From the result, we can see that the two pairs of double poles further separate, with the low-frequency double pole around 50 kHz. Compared with Fig. 18(b), this double pole is at a lower frequency, which indicates a longer settling time and the peaking also increases.

Fig. 20 shows a load step-up transient waveforms comparison for a 0.5–2 A step-up transient. For the $S_e = 3.5S_f = 2.2 \text{ mV}/\mu\text{s}$ case, the transient performance is very oscillatory due to small external ramp. As shown in Fig. 20(a), a very large overshoot

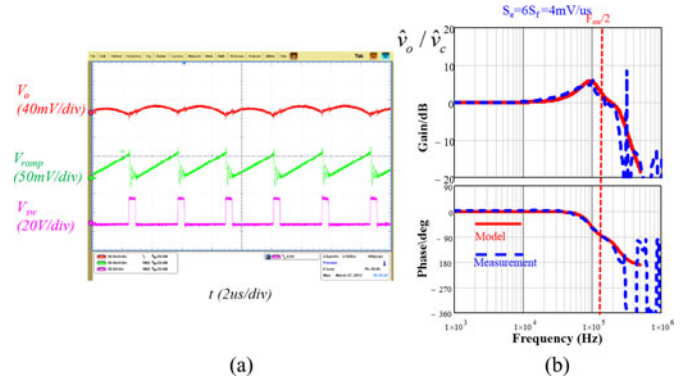


Fig. 21. (a) Steady-state waveform. (b) Small-signal control-to-output transfer function with $S_e = 6S_f = 4 \text{ mV}/\mu\text{s}$ and $D = 0.1$.

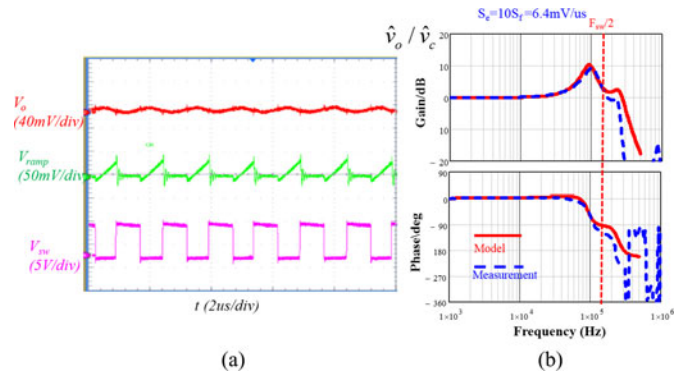


Fig. 22. (a) Steady-state waveform. (b) Small-signal control-to-output transfer function with $S_e = 10S_f = 6.4 \text{ mV}/\mu\text{s}$ and $D = 0.5$.

(around 140 mV) exhibits. To make a fair comparison between preferred case and the case using too large external ramp, the worst-case step-up transient performance is compared. For constant on-time control, the worst-case step-up transient point is at the end of the turn-off period: since the inductor current is at a minimum value at this point, if the step-up occurs, the largest capacitor discharge is required to provide the imbalance between the inductor current and load. For quantitative comparison, the undershoot voltage and settling time are compared as follows: For $S_e = 10S_f = 6.3 \text{ mV}/\mu\text{s}$, the undershoot is 60 mV and the settling time is $28 \mu\text{s}$ ($9T_{sw}$). For the $S_e = 30S_f = 19 \text{ mV}/\mu\text{s}$ case, the external ramp is so large that the low-frequency double pole increases the settling time to $48 \mu\text{s}$ ($16T_{sw}$) and the undershoot is 90 mV. The proposed external ramp provides the best dynamic performance among the three.

To verify the duty cycle effect on the dynamic performance, experimental results at different input voltages are shown as Figs. 21–23.

Fig. 21 shows the steady-state waveform and the small-signal control-to-output transfer function verification for the $D = 0.1$ case. In this case, the preferred external ramp is around $2S_{e_k} = 6S_f = 4 \text{ mV}/\mu\text{s}$.

Fig. 22 shows the steady-state waveform and the small-signal control-to-output transfer function verification for $D = 0.5$ case. In this case, the preferred external ramp is around $2S_{e_k} = 10S_f = 6.4 \text{ mV}/\mu\text{s}$. Note that in both cases, the external ramp

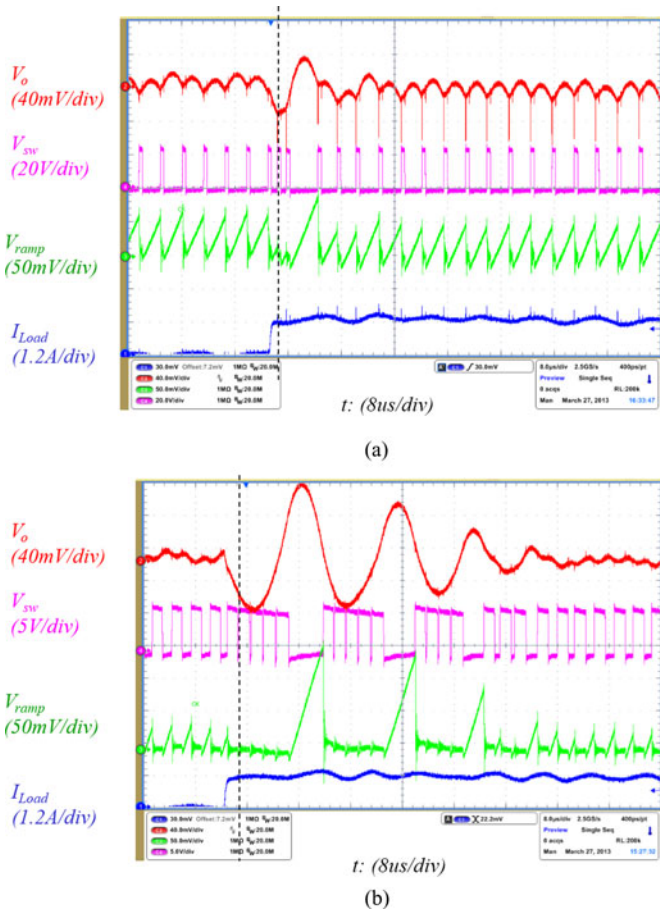


Fig. 23. Worst-case load step-up transient performance comparison for 0.5–2 A load step. (a) $D = 0.1$, $S_e = 6S_f = 4 \text{ mV}/\mu s$. (b) $D = 0.5$, $S_e = 10S_f = 6.4 \text{ mV}/\mu s$.

is designed around the best point to form a fair comparison. Figs. 21(b) and 22(b) show that for different duty cycle cases, the small-signal model agrees with the measurement results very well. The comparison shows that although the low-frequency pole position is similar in both cases, Fig. 22(b) shows a larger peaking, which indicates that for a large duty cycle case, the dynamic performance is worse. Fig. 23 compares the worst-case load step-up performance. For $D = 0.1$ case, the settling time is around $16 \mu s$ ($5T_{sw}$) and the undershoot voltage is around 35 mV. For the $D = 0.5$ case, however, the transient performance is much worse, with the settling time around $51 \mu s$ ($17T_{sw}$) and the undershoot voltage around 70 mV.

VI. CONCLUSION

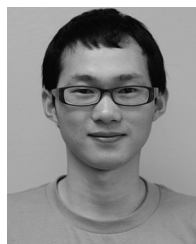
This paper investigates the characteristics of constant on-time V^2 control for ceramic capacitors with external ramp compensation by factorizing the small-signal model and identifying the pole-zero movements. The design of the external ramp is critical as it affects the position and damping of the double pole. In region I, both pairs of double poles locate at half of switching frequency. With increasing S_e , the two pairs of double poles will move toward each other and meet with each other at the key point. The damping of the circuit increases with increasing S_e in

this region. In region II, two pairs of double poles are separated, and the low-frequency double pole dominates the dynamic performance. The analysis shows that the recommended external ramp is around two times that of the value at key point, which is shown in (17). For a large duty cycle application, the dynamic performance is worse, and only external ramp cannot provide a well-damped performance. The small-signal simulation and experimental results verify the effect of the external ramp on the small-signal characteristic, and load transient experimental results verify the proposed design guideline for external ramp.

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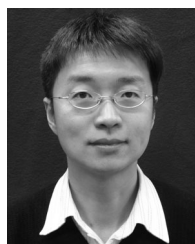


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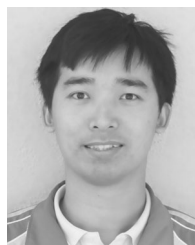
From 2003 to 2012, he served as an Associate Editor for the IEEE TRANSACTIONS ON POWER ELECTRONICS. From 2005 to 2010, he was the Industrial Power Converter Committee Technical Review Chair for the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS. For terms 2003–2006 and 2006–2009, he was also a member-at-large of the IEEE Power Electronics Society's Administrative Committee. He also received in 2005, 2006, and 2011 the Prize Paper Award in the IEEE TRANSACTIONS ON POWER ELECTRONICS, and in 2007, the Second Prize Paper Award at the IEEE Industry Application Annual Meeting.



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